

Online Appendix for:
*Trigger Versus Target Inflation:
Monetary Policy in a Dynamic Metastable Model of
Inflation*

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This Online Appendix to *Trigger Versus Target Inflation* provides details on several features of the model and data discussed in the paper. First it provides additional detail on the empirical properties of both the real US PCE inflation data and the simulated data from the computational model to what is discussed in the paper. It then provides more detail on the process used for estimating the stochastic kernel of the computational model. Next it provides a discussion of how variations in the price parameters of the monetary policy objective function effect the determination of optimal policy. Finally, details on some robustness testing of the model and links to code bases and data sets to enable replication of results are provided .

1 Empirical properties of the data, real and simulated

Figure 1 plots a time series of simulated observations of inflation that matches the length of the US PCE inflation series, comprised of the two data generating processes for the two sub-periods both of which match the length of the two sub-periods identified in the PCE chain-type inflation data. There are patterns in the time series that resemble those of Figure 2.1 in the paper. Sub-period 1 (observations 1 through 373) exhibits high volatility and variation in trend. Sub-period 2 is much less volatile and does not appear to trend up or down.

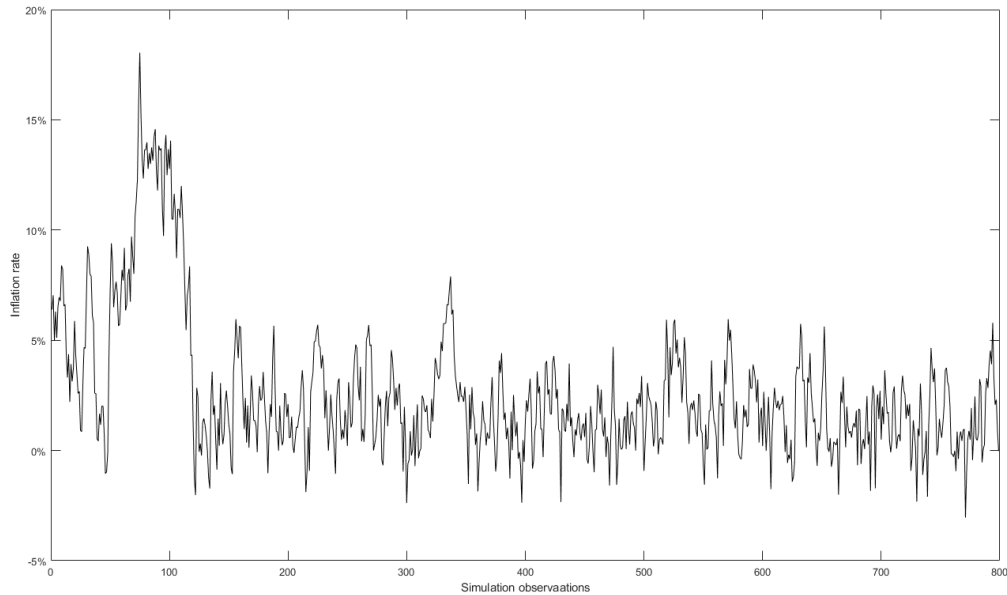


Figure 1: Simulated inflation over 2 regimes

1.1 Persistent relative price trends

A distinctive dynamic feature of the data is the well known persistence of relative price trends in the components of PCE. Figure 2 illustrates this feature of the data by plotting the inflation rate of four components of PCE – Durable goods, Services, Food and Energy – relative to the inflation rate of PCE. The plots in each of the four panels of Figure 2 are $\frac{1+\pi_i}{1+\pi} - 1$, where $i = \{Durable, Services, Food, Energy\}$. Evidently, persistent relative price trends are a feature of the PCE inflation process. The relative price trends of energy are particularly volatile and persistent in the later part of the time series. (This is reflected in the larger scale of the y-axis in Panel 4 relative to the other panels).

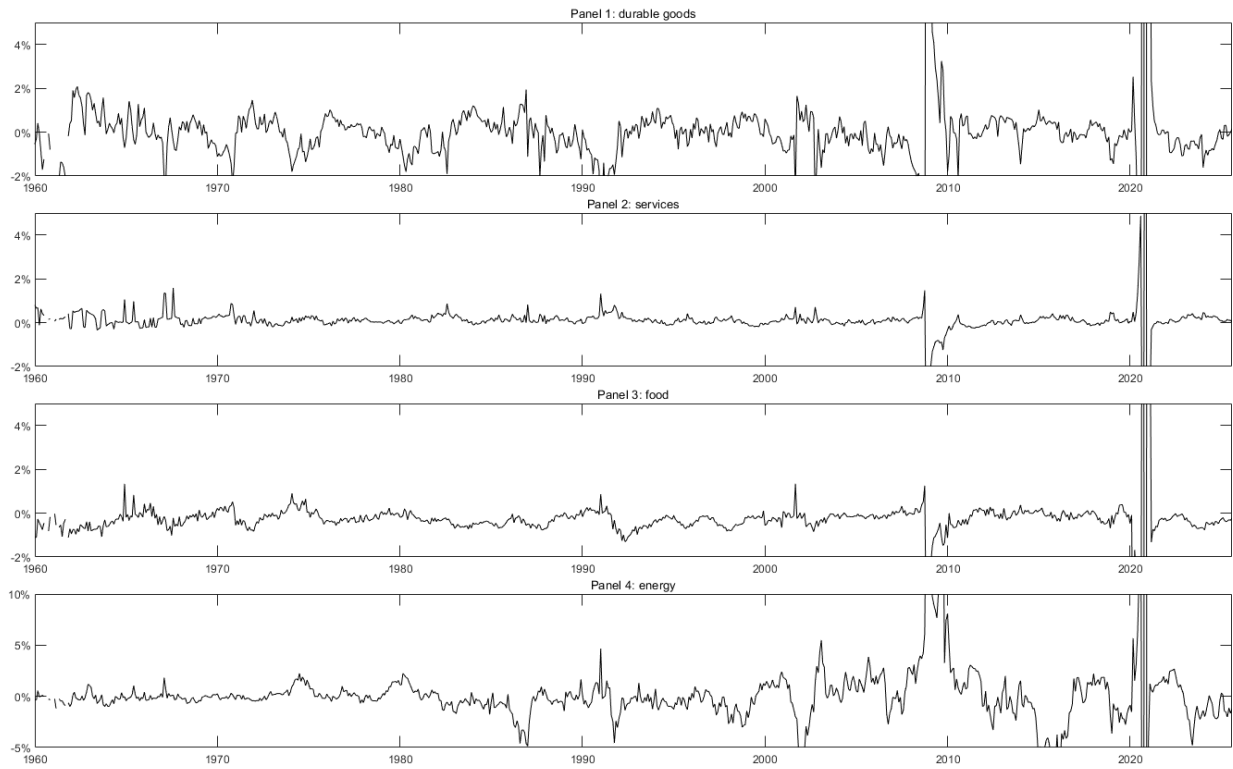


Figure 2: Relative price trends in components of PCE inflation

The base line simulation generates persistence of relative price trends as a natural feature of the dynamic process in the model. To emulate the real data a sub-set of 25 of the 100 prices being set by individual firms in the system is combined into an average grouping and compare that combined inflation rate to the overall. Figure 3 plots an illustrative sub-set of the average inflation of 25 prices in relation to the average inflation generated in each simulation period for the whole set of 100 prices in the system, $\frac{1+\pi(25)}{1+\pi(N=100)} - 1$. Aside from the significant exogenous interventions of the Financial Crisis and the COVID 19 Pandemic, the simulated sub set plotted in Figure 3 has a similar (but more volatile) pattern to that of PCE

sub components plotted in Figure 2, especially for the sub-period 1991-2026. Persistence in relative price inflations is clearly prevalent in both data sets.

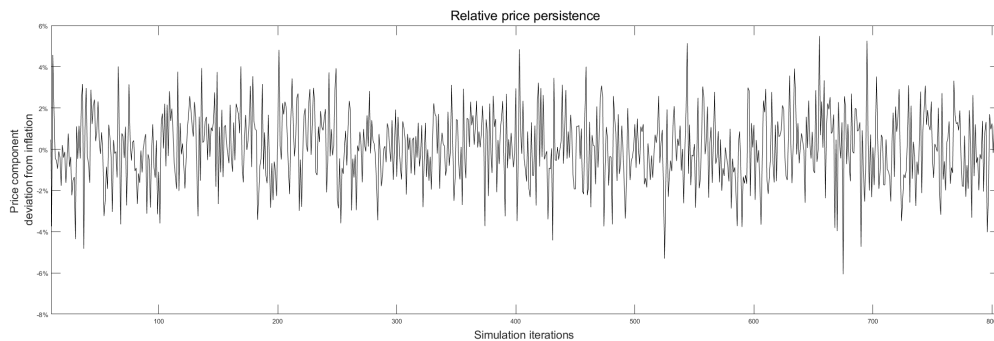


Figure 3: Relative price trends in components of simulation inflation

1.2 Regime switching

As noted in the paper, a test for evidence of a two state Markov regime-switching (MS) model is conducted on the real US PCE inflation series and set simulated data. Table 1 reports the estimation results for the whole period and the two sub-periods for the US PCE inflation series. Table 2 reports the estimation results for the simulation that matches the length of the whole series and two sub-periods in the real data.

Table 1: Markov Regime-switching, Chain-type PCE Inflation

Whole Period (1960-2025)					
	Coefficient	Estimate	Standard Error		
Regime 1	α_1^{MS}	0.0202	0.0274		
	β_1^{MS}	1.0039***	0.0050		
Regime 2	α_2^{MS}	0.0562***	0.0115		
	β_2^{MS}	0.9775***	0.0033		
Regime St. Dev.	σ_1^{MS}	0.4399	—		
	σ_2^{MS}	0.1797	—		
Transition Probability	$p_{1,1}^{MS}$	0.885	—		
	$p_{2,2}^{MS}$	0.960	—		
Sub-Period 1 (1960-1990)			Sub-Period 2 (1991-2025)		
	Coefficient	Estimate	Standard Error	Estimate	Standard Error
Regime 1	α_1^{MS}	0.0989***	0.0196	0.0119	0.0407
	β_1^{MS}	0.9610***	0.0043	0.9990***	0.0124
Regime 2	α_2^{MS}	0.3390***	0.0339	0.1205***	0.0205
	β_2^{MS}	0.9895***	0.0046	0.9397***	0.0090
Regime St. Dev.	σ_1^{MS}	0.1955	—	0.4942	—
	σ_2^{MS}	0.2413	—	0.1784	—
Transition Probability	$p_{1,1}^{MS}$	0.971	—	0.850	—
	$p_{2,2}^{MS}$	0.885	—	0.955	—

Notes: ** and *** indicate significance at the 5% and 1% level, respectively.

Table 1 shows that U.S. inflation between 1960 and 2025 displays clear evidence of two distinct regimes. In both regimes, the persistence of inflation, which is measured by the autoregressive coefficient, is similar (1.0039 in Regime 1 and 0.9775 in Regime 2). However, the intercepts differ. The intercept in Regime 1 is statistically insignificant at 0.0202, while that in Regime 2 is positive and significant at 0.0562.

These estimates imply that, if last month's inflation rate was 2%, the expected current inflation would be approximately 2.028% in regime 1 and 2.011% in regime 2. When the previous month's inflation is to 4%, the expected inflation diverges between the two regimes, increasing slightly to 4.036% in Regime 1, but declining slightly to 3.956% in Regime 2. Also, regime 1 exhibits substantially higher volatility, with a standard deviation of 0.4399 compared with 0.1797 in Regime 2. Regime 1 also displays a slightly lower transition probability.

The two sub periods display evidence of two regimes but they differ from each other. Obviously the volatility and the insignificance on the intercept estimate in regime 1 for the whole period is largely driven by the process occurring in sub period 2.

The simulation where matches the number of periods to the real data series. The results obtained indicate two regimes in the simulated data for the whole period and for each sub period. Again the sub periods differ from each other and the transition probabilities indicate strong persistence in all cases shown in Table 2.

Table 2: Markov Regime-switching, Simulation

Whole Period (simulated 1960-2025)					
	Coefficient	Estimate	Standard Error		
Regime 1	α_1^{MS}	0.0066***	0.0006		
	β_1^{MS}	0.5067***	0.0287		
Regime 2	α_2^{MS}	0.0056	0.0009		
	β_2^{MS}	0.9215***	0.0122		
Regime	σ_1^{MS}	0.0132	–		
St. Dev.	σ_2^{MS}	0.0151	–		
Transition	$p_{1,1}^{MS}$	0.981	–		
Probability	$p_{2,2}^{MS}$	0.947	–		
Sub-Period 1 (simulated 1960-1990)			Sub-Period 2 (simulated 1991-2025)		
	Coefficient	Estimate	Standard Error	Estimate	Standard Error
Regime 1	α_1^{MS}	0.0064***	0.0010	0.0092***	0.0010
	β_1^{MS}	0.6148***	0.0380	0.6536***	0.0356
Regime 2	α_2^{MS}	0.0054***	0.0015	0.0061***	0.0007
	β_2^{MS}	0.9324***	0.0177	0.3680***	0.0426
Regime	σ_1^{MS}	0.0131	–	0.0133	–
St. Dev.	σ_2^{MS}	0.0145	–	0.0087	–
Transition	$p_{1,1}^{MS}$	0.991	–	0.879	–
Probability	$p_{2,2}^{MS}$	0.978	–	0.931	–

Notes: ** and *** indicate significance at the 5% and 1% level, respectively.

1.3 Unit roots

The "augmented" part of the (ADF) test involves adding lagged difference terms to a regression model to remove the effects of autocorrelation. The ADF test is used to examine the data over the whole period and the two sub-periods identified in Section 2 of the paper.

Table 3 presents results that mostly detect stationarity in the data from applying various formulations of the ADF test to the PCE inflation time series in the whole period and the two sub-periods. The ADF test is run with four lags on its the straightforward auto-regressive formulation, a formulation with drift but trend stationarity, and formulation with trend but no drift. Each

Table 3: ADF Test, US chain-type PCE inflation

Sample Size	Crit. val. α	Auto Regressive* lag={0, 1, 2, 3, 4}	Trend Stationary lag={0, 1, 2, 3, 4}	No Drift lag={0, 1, 2, 3, 4}
Whole period (1960-2026)	0.10	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 1, 1, 1, 1
	0.05	1, 1, 1, 1, 0	1, 1, 1, 1, 1	1, 1, 1, 1, 1
	0.01	1, 0, 0, 0, 0	1, 1, 1, 1, 1	1, 1, 1, 1, 1
Sub-period 1 (1960-1990)	0.10	0, 0, 0, 0, 0	1, 0, 0, 0, 0	1, 1, 1, 0, 1
	0.05	0, 0, 0, 0, 0	1, 0, 0, 0, 0	1, 0, 0, 0, 0
	0.01	0, 0, 0, 0, 0	0, 0, 0, 0, 0	1, 0, 0, 0, 0
Sub-period 2 (1991-2026)	0.10	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 1, 1, 1, 1
	0.05	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 1, 1, 1, 1
	0.01	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 1, 1, 1, 1

* Default setting for the ADF test, no drift, no trend.

1 represents rejection of the null hypothesis of a unit root.

0 represents failure to reject the null and implies the possibility of a unit root.

formulation is tested with a critical value of 0.10, 0.05 and 0.01. Table 3 shows that the whole period shows limited evidence of a unit root. Evidence of a unit root is rejected across the board in the second sub-period (1991-2026). However, strong evidence of a unit root is detected in the first sub-period (1960-1990). Clearly, the limited evidence of a unit root in the whole period is being driven by the evidence of a unit root in the first sub-period.

As in the real data, the various test for unit roots can be conducted on the whole simulated time series and the sub-periods representing the two regimes identified there. Table 4 show the results of the same set of tests for the simulated data.

Again evidence for unit roots in the simulated sub-period 1 is detected. Weaker evidence for unit roots in the whole simulated time series and a clear rejection of unit roots in simulated sub-period 2 is also detected.¹ The simulation allows for the production of as much data as required to test the robustness of the unit root results by extending the time series in both sub-periods

¹ The results here are for a specific data run. More generally we find varying evidence for the unit root in simulated sub-period 1. Resampling for 373 observation over a long simulation run the simulated evidence of a unit root shows up occasionally, not overwhelmingly.

Table 4: Augmented Dickey-Fuller (ADF) Test, simulated inflation

Sample Size	Crit. val. α	Auto Regressive* lag={0, 1, 2, 3, 4}	Trend Stationary lag={0, 1, 2, 3, 4}	No Drift lag={0, 1, 2, 3, 4}
Sim whole period	0.10	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 1, 0, 0, 0
	0.05	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 0, 0, 0, 0
	0.01	0, 0, 0, 0, 0	1, 0, 0, 0, 0	0, 0, 0, 0, 0
Sim Sub-period 1	0.10	0, 0, 0, 0, 0	0, 0, 0, 0, 0	0, 0, 0, 0, 0
	0.05	0, 0, 0, 0, 0	0, 0, 0, 0, 0	0, 0, 0, 0, 0
	0.01	0, 0, 0, 0, 0	0, 0, 0, 0, 0	0, 0, 0, 0, 0
Sim Sub-period 2	0.10	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 1, 1, 1, 1
	0.05	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 1, 1, 1, 1
	0.01	1, 1, 1, 1, 1	1, 1, 1, 1, 1	1, 1, 1, 1, 1

* Default setting for the ADF test, no drift, no trend.

1 represents rejection of the null hypothesis of a unit root.

0 represents failure to reject the null and implies the possibility of a unit root.

and, thus, for the overall time series. When the times series for sub-period 1 is extended to 5,000 observations all traces of a unit root disappear. Perhaps less surprising is that there is no evidence of unit root in sub-period 2 or the whole times series when the number of observations is increased. Stationarity in the simulation results should not be surprising given that the kernel estimation procedure discussed in Section 3 of the paper and in Section 2 below requires that the inflation generating process be stationary. What may be of some interest is that evidence of a unit root exhibited by the short time series available in the US PCE data in sub-period 1 may be spurious. If a longer time series of the same data generating process for sub-period 1 were available, perhaps evidence of a unit root may also disappear.

2 Estimating the the stochastic kernel density

Because prices increase without bound in the presence of persistent positive inflation, there is a problem of running into computational size limits in long simulations that produce the price

series. Following [Perla, Sargent and Stachurski's Quantitative Economics with Julia](#) we estimate the stationary density of inflation for the process by iterating over a given stochastic kernel that determines prices. Here we establish that iterating over the dynamic process of the pricing decisions of firms produces a stationary inflation rate by looking at relatively long simulations of the price sequence produced by such a process.

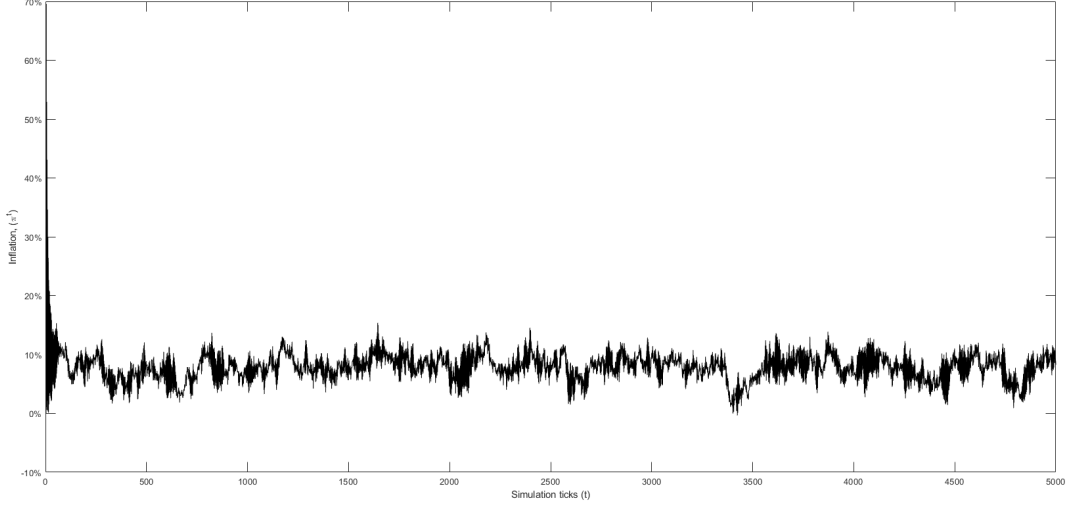


Figure 4: Inflation in the pricing kernel

Figure 4 presents a series of inflation derived from an illustrative simulation run of the price setting process where prices are allowed to increase throughout the run. Initially all firm prices are set to 1. The baseline parameterization reported in the text, with the exception that full information is available to agents in the system, $\gamma = 1$ and $\eta = 1$ is used. For this illustration, the process quickly settles into stable inflation with a mean of 7.85% and standard deviation of 2.25%. The price level at the end of 5000 simulation iterations is $1.98 \cdot 10^{176}$.

Figure 4 reveals three important observations.

1. The kernel produces stable inflation over the simulation run.
2. The kernel takes several iterations to settle into a stable inflation rate.
3. Prices become very large very rapidly in the presence of persistent positive inflation.

These observations have implications for modelling the system. Foremost among them is the simulation reported in Figure 4 is for only 5000 iterations. During it, average price increases from an initial value of 1 to a huge number $1.98 \cdot 10^{176}$. To establish convergence and estimate the stationary density in the baseline simulation requires many times more iterations for every monetary policy considered. Therefore, simply running a simulation for each monetary policy, allowing prices to increase through the number of iterations required to establish convergence is not feasible. Instead, for each value of inflation, the pricing kernel is run until inflation stabilizes and the value of inflation at that point forms one observation in the series. This process is iterated, reseeding the kernel with initial prices taken from the previous kernel each time. Because of the lag structure in the formulations of expected inflation, expected wages and observed prices the process to establish p_n^{t-1} and p_n^t in order to calculate inflation is of length 3. Each convergence simulation iterates over the kernel starting with a set of prices and wages where each is equal to 1 for each firm. But as the simulation iterates over the kernel, it is reseeded with individual firm wages, w_n^{t-1} and observations of prices p_n^{t-1} from the previous iteration.² This produces a stationary density of inflation without prices exploding.

Established the procedure to generate inflation without hitting computational limits of an ever increasing price level, enables examination of the density properties of the state variable. The procedures provided in Perla, Sargent and Stachurski's *Quantitative Economics with Julia* are followed here to specify the Markov operator, corresponding to z . Let D be the set of all densities on $\mathbf{S}$, and let Z be the operator from D to itself that takes density ψ and sends it into new density ψZ , where the latter is defined by

$$(\psi Z)(y) = \int z(x, y) \psi(x) dx."$$

This enables the specification of the density of the stochastic kernel as

$$\psi^{t+1}(y) = (\psi^t Z)(y) \forall y,$$

² Convergence simulations have also been run, reseeding the prices and wages with 1's each iteration and similar results in terms of stationary inflation series were obtained. However, reseeding the kernel in this way is computationally inefficient because as observed in Figure 4 the kernel takes several iterations to stabilize on the stationary inflation rate. Reseeding with prices from the previous iteration produces stabilized inflation much more rapidly.

which can be written as

$$\psi^{t+1} = (\psi^t Z)$$

by dropping the y and understanding “=” to mean equality of functions. As noted by Perla, Sargent and Stachurski, this means that specifying a distribution ψ^0 and then iterating with Z enables one to obtain the entire sequence of future distributions. This is done here, using what Perla, Sargent and Stachurski call the naive approach, by iterating with Z to form the histogram which is the estimate of the stationary density of the process at convergence.

Following Bekar et. al (2025) and Perla, Sargent and Stachurski a convergence criteria is created for the estimate of the stationary density. A frequency density of the inflation process in the simulation model is calculated as a histogram over a range of possible values from $(-2.00\% \cdots 33.00\%)$, binning into intervals of size 0.35% as

$$\mathbf{F}\boldsymbol{\pi}(n\pi_0^t, n\pi_1^t, \dots, n\pi_N^t),$$

where $n\pi_j^t$ is the number of instances in the first t periods of the simulation in which there were observations of π fell into one of the 100 intervals. The simulation is run in blocks of 20,000 iterations. At the end of each block, $\mathbf{F}\boldsymbol{\pi}$ is used to calculate the relative frequency distribution of violations $\mathbf{RF}\boldsymbol{\pi}^{\hat{n}} = (r\pi_0^{\hat{n}}, \dots, r\pi_N^{\hat{n}})$, where $r\pi_j^{\hat{n}}$ is the proportion of periods in which there were exactly j observations of π in each of the j intervals over the first $50000 \cdot \hat{n}$ periods of the simulation. As $\hat{n}$ approaches infinity $\mathbf{RF}\boldsymbol{\pi}^{\hat{n}}$ approaches the stationary density. How large must $\hat{n}$ be to ensure that $\mathbf{RF}\boldsymbol{\pi}^{\hat{n}}$ is a *good* estimate of the stationary density? To answer this we need a convergence criterion.

At the end of each simulation block, for all $\hat{n} \geq 2$, the following test statistic is calculated:

$$TEST^{\hat{n}} = \sum_{j=0}^{j=N} |r\pi_j^{\hat{n}} - r\pi_j^{\hat{n}-1}|$$

$TEST^{\hat{n}}$ is a measure, decreasing in $\hat{n}$, of the *distance* between $\mathbf{RF}\boldsymbol{\pi}^{\hat{n}}$ and $\mathbf{RF}\boldsymbol{\pi}^{\hat{n}-1}$. $TEST^{\hat{n}}$ does not decline monotonically, which complicates any test of convergence. $TEST^{\hat{n}} \leq 0.001$ for 5 consecutive blocks of $\hat{n}$ is the convergence criteria used. Denoting the minimum $\hat{n}$ for which the convergence criterion is satisfied by $\hat{n}^*$ yields an approximation of the stationary density, $\mathbf{RF}\boldsymbol{\pi}^{\hat{n}^*} \sim$

stationary density.

3 Monetary policy objective function

In Section 5 of the paper it is observed that the optimal Two-Bin monetary policy achieves significant efficiency in price stability and employment maximization relative to a static policy and was very nearly as good at minimizing costs as a three-bin policy. The approximately optimal Two-Bin policy for the baseline simulation is $(\pi^*, i_1^*, i_2^*) = (4.00\%, 0.25\%, 7.50\%)$. As is noted in the paper, the optimal policy is a function of the relative cost of the monetary policy interest rate that of inflation. Changing that ratio and assessing whether the optimal Two-Bin policy for the baseline is dominated by alternatives when the price of inflation is larger relative to the price of the interest rate demonstrates that point.

For prices of $\lambda_1 = 1$ and $\lambda_2 = 10$, the baseline optimal policy costs 0.1865 and the policy $(\pi, i_1, i_2) = (4.00\%, 1.00\%, 7.50\%)$ costs 0.1778, or 4.7% less, on average. For prices of $\lambda_1 = 1$ and $\lambda_2 = 20$, the baseline optimal policy costs 0.3661 and the policy $(\pi, i_1, i_2) = (4.00\%, 1.25\%, 7.50\%)$ costs 0.3427, or 6.4% less, on average. In both of these test cases the policy $(\pi, i_1, i_2) = (4.00\%, 0.00\%, 7.50\%)$ cost more than the optimal baseline Two-Bin policy.³

As noted in the text it is an empirical question as to what the price of inflation in relation to the price of the policy interest rate is. But this framework offers some interesting possibilities for analyzing what the monetary policy implications are once that empirical determination is made.

4 Robustness and replication

4.1 Variation in σ

Figure 5 shows the plot of expected inflation in the stationary density for the baseline (solid line) and a new value of $\sigma = 0.09$ (dashed line). The decrease in σ shifts the position of the transition regime slightly and makes it more abrupt. The focal values of LA and HA are reduced slightly.

³ The two policies that exhibit lower costs at the new relative price of inflation are approximately optimal ones. The search algorithm that finds the approximately optimal policies given a parameterization is computationally expensive, and without more empirical information about what the price of the policy interest rate, in relation to the the price of inflation should be to guide it, an exhaustive search in the parameter space seems infeasible.

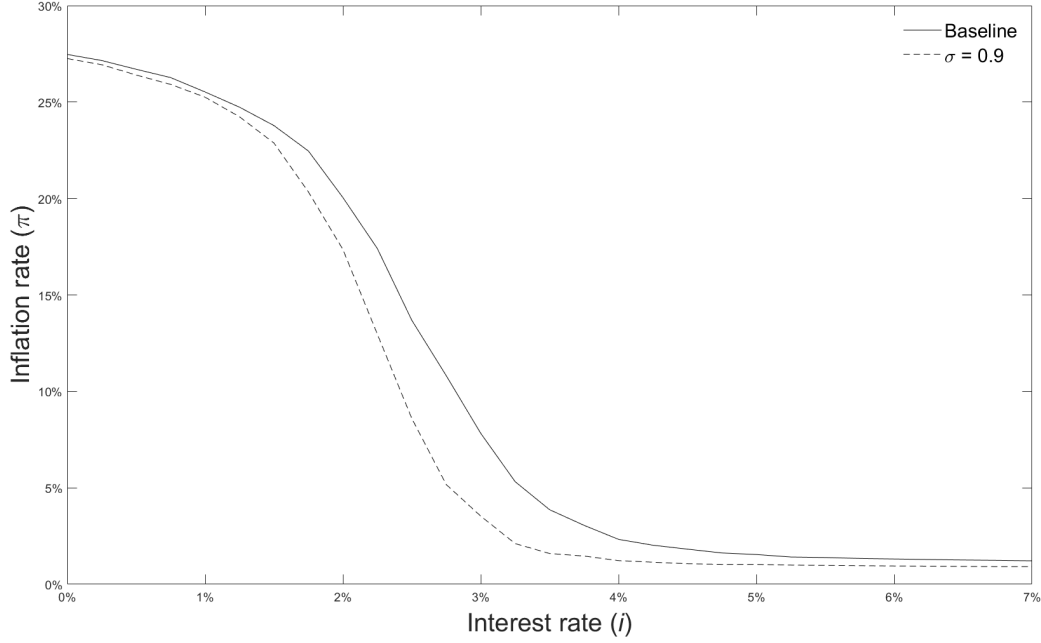


Figure 5: Expected inflation for different parameterizations

The increase in the abruptness (steepness) of the transition regime reflects an increase in the persistence of the q-attractors.

4.2 Variation in N

Figure 6 shows the plot of expected inflation in the stationary density for the baseline (solid line) and the new value of $N = 80$ (dashed line). A decrease in the number of firms slightly alters the transition regime, implying a lower focal value of HA. This is somewhat intuitive since inflation for the system is the average average inflation for the whole system when a smaller number of firms are either not increasing prices or increasing them very little the LA stays in the relative same position. But because overall inflation is from all firms in the system, if all of firms are increasing prices and potentially by a lot (as is the case in HA), the overall inflation of the system when in HA will occur at a lower focal value for a lower total number of firms. The change also flattens (or stretches out) the transition regime.

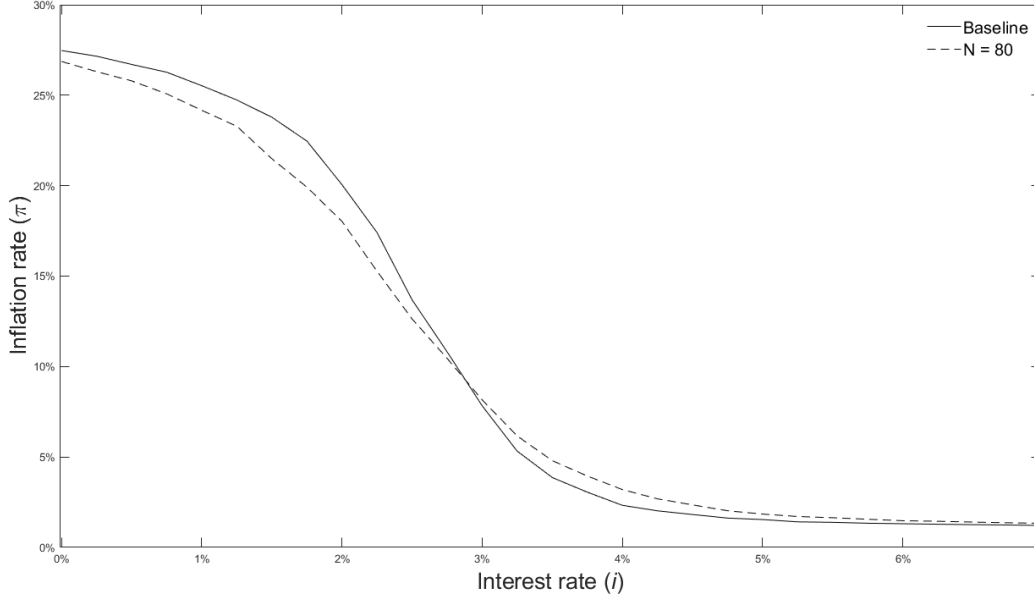


Figure 6: Expected inflation for different parameterizations

4.3 Calibration

Figure 7 shows the plot of expected inflation in the stationary density for the baseline (solid line) and the same plot for different values of a collection of parameters (dashed line). $\theta = 1.15$, $\sigma = 0.08$ and $\psi = 0.005$ is contrasted with the baseline. This illustrates that the model's transition regime is reproducible from a variety of different parameter combinations, over a range of different inflation values with different focal values for the q-attractors. The crude calibration done in the paper enables the limitation of the number of variations considered. But obviously there are still vast possibilities even within the calibration's confines.

4.4 Code for replicating figures and results

Figure 4.1 in the paper, presents time series data on inflation and information from five simulations of the model. $i = (1.0\%, 2.25\%, 2.50\%, 4.25\%, 5.5\%)$. The left-hand column shows five pairs of panels for the time series of inflation and information (measured as K_e^t) for each monetary policy over 5000 periods. The middle column of panels presents the corresponding frequency densities of values of inflation, presented in bin intervals of 0.01. The right-hand column of panels presents

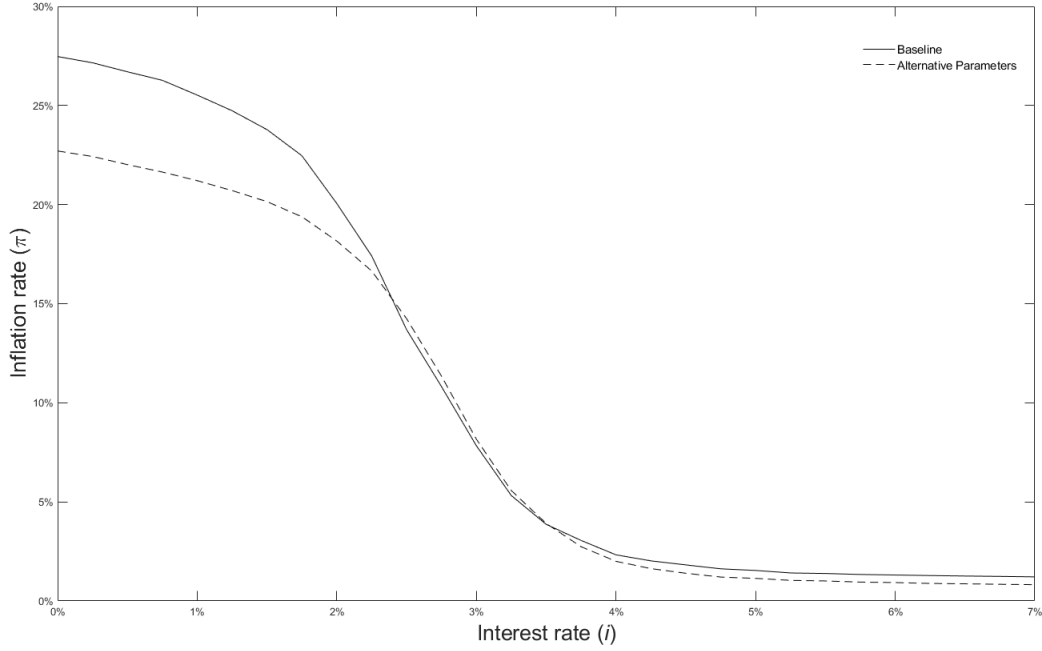


Figure 7: Expected inflation for different parameterizations

the autocorrelation coefficients of inflation for lags 1, 2 and 3.

Code for producing Figure 4.1 from Section 4 can be found here:

- [Figure 4.1](#)

Figure 4.2 presents the average value of inflation in the estimated stationary density at the convergence criteria for each value of $i = (0.00\%, 0.25\%, \dots, 7.00\%)$. The data in this plot is saved as Matlab data file **mibl26.mat** and is used in several other code bases.

Code for producing Figure 4.2 from Section 4 can be found here:

- [Figure 4.2](#)
- [mibl26.mat](#)

Figure 4.3 presents the q-attractors for $i = 2.50\%$. It can be run in two ways:

1. run the code for at least 10M simulation ticks. The recommend procedure is as follows:

- at line 14 in the code base set the size of $Block < 40K$ to ensure the kernel does not crash.
 - toggle off the load command on line 50 and the statement on line 51 using "%"
 - toggle on the command on line 56 (i.e., ensure it is not preceded by "%")
 - at line 200 in the code base set the conditional statement "if $bb > X - 1$ " such that $X \cdot Block = 10M$
2. The much faster alternative is to use the data file `iin26_1.mat` already created. The recommended procedure is as follows:
- toggle on the load command on line 50 and the statement on (i.e., ensure each is not preceded by "%")
 - toggle off the command on line 56 using "%"

Figure 4.4 presents the q-attractor for $i = 5.50\%$. It can be run in two ways:

1. run the code for at least 45M simulation ticks. The recommend procedure is as follows:
- at line 14 in the code base set the size of $Block < 40K$ to ensure the kernel does not crash.
 - toggle off the load command on line 50 and the statement on line 51 using "%"
 - toggle on the command on line 56 (i.e., ensure it is not preceded by "%")
 - at line 200 in the code base set the conditional statement "if $bb > X - 1$ " such that $X \cdot Block = 45M$
2. The much faster alternative is to use the data file `iin26_2.mat` already created. The recommended procedure is as follows:
- toggle on the load command on line 50 and the statement on (i.e., ensure each is not preceded by "%")
 - toggle off the command on line 56 using "%"

Code for producing Figure 4.3 from Section 4 and the data file `iin26_1.mat` can be found here:

- [Figure 4.4](#)
- [iin26 2.mat](#)

Figure 4.5 presents three panels of plots showing the persistence and time spent in the q -attractors.

Code for producing Figure 4.5 from Section 4 and the data file it uses **`mibl26.mat`** can be found here:

- [Figure 4.2](#)
- [mibl26.mat](#)

Figure 5.1 presents the plot of the cost estimated from the stationary density for the baseline.

Code for producing Figure 5.1 from Section 4 and the data file it uses **`mibl26.mat`** can be found here:

- [Figure 5.1](#)
- [mibl26.mat](#)

Figure 5.2 presents the q -attractor for the optimal Two-Bin policy in Section 5. It can be run in two ways:

1. run the code for at least 45M simulation ticks. The recommend procedure is as follows:
 - at line 14 in the code base set the size of $Block < 40K$ to ensure the kernel does not crash.
 - toggle off the load command on line 52 and the statement on line 53 using `"%"`
 - toggle on the command on line 57 (i.e., ensure it is not preceded by `"%"`)
 - at line 200 in the code base set the conditional statement `"if  $bb > X - 1$ "` such that $X \cdot Block = 60M$

2. The much faster alternative is to use the data file `iin2b26.mat` already created. The recommended procedure is as follows:

- toggle on the load command on line 52 and the statement on line 53 (i.e., ensure each is not preceded by `%`)
- toggle off the command on line 57 using `%`

Code for producing Figure 5.2 from Section 4 and the data file `iin2b26.mat` can be found here:

- [Figure 5.2](#)
- [iin2b26.mat](#)

Figures 2.2 and 6.1 are produced using Matlab code bases and data files that can be found here:

- [Figure 2.2](#)
- [PCE.mat](#)
- [Figure 6.1](#)
- [sim26.mat](#)
- [simsub1.mat](#)
- [simsub2.mat](#)

The Matlab codes bases for the search routines used to fine the approximately optimal Two-Bin and Three-Bin policies can be found here:

- [Two-Bin Search](#)
- [Three-Bin Search](#)